

RYAN BARRY

Game Developer | C#, Unreal Engine, MonoGame and Unity Specialist Dublin/Sligo, Ireland | 089 244 7817 | [Email](#) | [LinkedIn](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Award-winning third-year Games Development student targeting a First Class Honours at ATU Sligo with a strong foundation in C# programming and engine architecture. Built a custom 2D game engine using the **MonoGame framework** demonstrating a deep understanding of memory management, rendering pipelines, and systems design. Possesses 5+ years of management experience in high-volume hospitality, bringing professional maturity, leadership, and agile workflow management skills to development teams.

TECHNICAL SKILLS

- **Languages:** C# (Proficient), C++, XAML, HTML, Python, SQL.
- **Engines:** MonoGame (Framework), Unity 2D/3D, Unreal, WPF.
- **Core Concepts:** Object-Oriented Programming (OOP), Game Loop Architecture, Systems Design, Sprite Batching & Rendering, Collision Algorithm, Async & Await.
- **Tools:** Visual Studio, Git / GitHub, Trello (Project Management), Photoshop, Maya, VS code.

AWARDS

- **Best Use of AI Algorithms** Sponsored by Electronic Arts Games Fleadh 2026

PROJECTS

[Pale Roots](#) | Solo Developer | C#, MonoGame, Custom Game Engine | Dec 2025 – March 2026 *An action packed Roguelike with high risk high reward leveling.*

- **Award-Winning Architecture:** Engineered a custom 2D engine from scratch, bypassing commercial engines to master low-level architecture; won the EA-sponsored "Best Use of AI" at Games Fleadh 2026.
- **Custom AI Systems:** Designed a Stack-Based Finite State Machine and modular IAISate architecture, allowing NPCs to dynamically evaluate surroundings and seamlessly switch between Wander, Chase, and Combat states.
- **Targeting & Physics:** Programmed dynamic spatial targeting algorithms to prevent unrealistic swarming, and implemented continuous momentum physics for fluid, non-glitchy movement.
- **System Decoupling:** Built an Event-Driven Combat System that broadcasts C# Action events, completely decoupling combat logic from the UI and state managers.

Split-Shift | Programmer | Unity, C# | Sept 2026 – Present

An open-world arcade racing game focused on high-stakes street races, faction missions, and dynamic vehicular combat.

- **Advanced Racer AI:** Engineered custom AI for rival racers utilizing a Waypoint/Spline system to outline tracks, bypassing standard NavMesh to eliminate robotic movement. Integrated advanced Steering Behaviors (Seek, Flee, Avoidance) enabling AI cars to dynamically overtake opponents and navigate around sudden obstacles while adhering to the spline path.
- **Police & Gang Pursuit FSM:** Implemented robust Finite State Machines (FSM) in C# to manage police vehicles and gang-affiliated NPCs. This state-driven architecture dynamically governs high-speed reactions to player vehicles, enabling sophisticated pursuit, interception, and evasion tactics.

Dying Fire | Solo Developer | January 2026 – June 2026 A hybrid Visual Novel / RPG Engine built entirely in WPF & XAML.

- **System Architecture:** Manipulated the WPF UI framework to function as a game loop, utilizing the **State Pattern** for scene management alongside strict **MVVM**.
- **Data Handling:** Utilised **SQLite** to instantiate and populate a **Database** which the **WPF** app read from to create the objects in the Game World and Inventory.
- **OOP Design:** Heavily utilized **Polymorphism** and **Interfaces** to decouple game entities from UI elements.

EDUCATION

Bachelor of Applied Science in Computer Games Development ATU Sligo | Sept 2023 – Present (Expected 2027)

- **Key Modules:** Object-Oriented Programming, Game Engine Architecture, Data Structures & Algorithms, Mathematics for Games.
- **Current Standing:** On track for First Class Honours.

Leaving Certificate Ardgillan Community College, Dublin | Achieved 520 Points

WORK EXPERIENCE

Manager & Head Bartender Lillies Cocktail Bar & Tannery Lane | Sligo | 2024 – Present

- **Leadership:** Managed a team of staff, handling rostering, training, and conflict resolution in a high-pressure environment.
- **Project Management:** Oversaw inventory control and stock management, requiring meticulous attention to detail similar to resource management in development.
- **Professional Growth:** Promoted to Head Bartender at age 17, demonstrating reliability and the ability to master complex skills rapidly.

GRADES

Current GPA: 1:1 First Class Honours

Module	Grade
3D Game Programming	95
2D Game Programming	78
Software Quality and Testing	81
Database Technology	88
Mathematics 4	76
Data Structures and Algorithms	72
Object Oriented Development	71